ÉCOLE MILITAIRE POLYTECHNIQUE

Department of Computer Science – Data Mining Course (`CS-504`)

Mini-Project Proposal: Heart Disease Risk Prediction

Advanced Machine Learning & Predictive Data Mining Study

Student Contributor Matrix (Simulated):

Student AISD05 Student SIAD19 Student RSD20

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Abstract

This mini-project proposal outlines a comprehensive Data Mining investigation on the Kaggle dataset `uditjain13/heart-disease-risk-2026` ($N = 9,000$). The project aims to develop a leakage-free machine learning classification pipeline, execute 6 non-parametric statistical hypothesis tests ($\alpha = 0.05$) with Benjamini-Hochberg FDR corrections, benchmark 6 supervised algorithms (Zero-R, KNN, Decision Tree, Random Forest, Naive Bayes, SVM), evaluate multi-perspective feature importances, and deploy an interactive 9-page Streamlit web dashboard.

1 Project Background & Motivation

Cardiovascular diseases account for 17.9 million annual deaths globally. Early detection using non-invasive clinical biomarkers offers immense potential for primary care decision support. Machine learning techniques allow extracting subtle non-linear interactions across physiological stress tests, blood chemistry, and lifestyle habits.

2 Core Research Objectives

1. Perform complete data quality audit ($N = 9,000$, 25 features, zero missing values, zero duplicates).
2. Construct leakage-free scikit-learn `ColumnTransformer` preprocessing pipelines.
3. Execute a 6-hypothesis statistical testing suite with Benjamini-Hochberg FDR corrections and effect size metrics (Cramér's V, Cohen's d, Odds Ratios).
4. Benchmark 6 supervised classifiers using 5-Fold Stratified Cross-Validation and hyperparameter grid search.
5. Deploy an interactive 9-page Streamlit web application featuring a 7-tier clinical risk stratification scale (0% to 100%).

3 Expected Deliverables

- Production-ready Python codebase (`src/` and `scripts/`).
- Trained model artifacts (`best_model.joblib`, `preprocessing_pipeline.joblib`).
- Interactive 9-page Streamlit web dashboard (`src/app.py`).
- Complete scientific LaTeX report and PDF document (`docs/Rapport_MiniProject.pdf`).
- Reproducible GitHub repository with simulated feature-branch workflow (`main`, `develop`, `feature/*`).

4 Timeline & Milestone Schedule

Table 1: Mini-Project Implementation Schedule

Phase / Milestone	**Deliverable / Activity**	**Assigned Student**
Phase 1: Audit & EDA	Dataset acquisition, quality audit, hypothesis testing	Student AISD05
Phase 2: Modeling & CV	Preprocessing pipeline, 6 ML classifiers, GridSearch	Student SIAD19
Phase 3: Web App & Report	Streamlit app, 7-tier predictor, LaTeX PDF report	Student RSD20